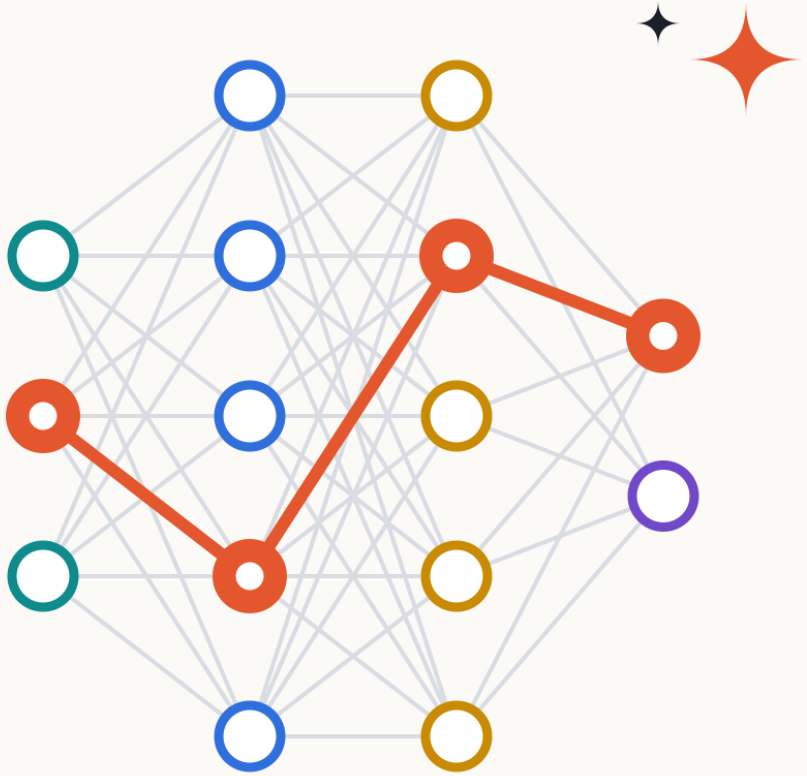
CS313

# Artificial Intelligence

## From Problems to Learning



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First Semester 2026/2027

Problem before terminology · Tree before branch · Reason before recipe

PHASE 1 · FOUNDATIONS

# 01

# AI Around Us: Spot the Intelligence

*What makes a system intelligent?*


01 AI Around Us
02 Intelligent Agents
03 Problem Formulation
04 Search
05 Searching Smarter
06 Constraints
07 Knowledge
08 Reasoning
09 Uncertainty
10 Learning from Data
11 Modern AI
12 Bringing It Together

# How many AI systems did you use today?



Face Unlock



Social apps



Google Maps



Music and video



Spam filter



Web search



ChatGPT

# Why did we call these systems AI?

*What exactly did you see that made you say: "This system is intelligent"?*



# Example 1: the calculator



$$125 \times 73$$

$$= 9125$$

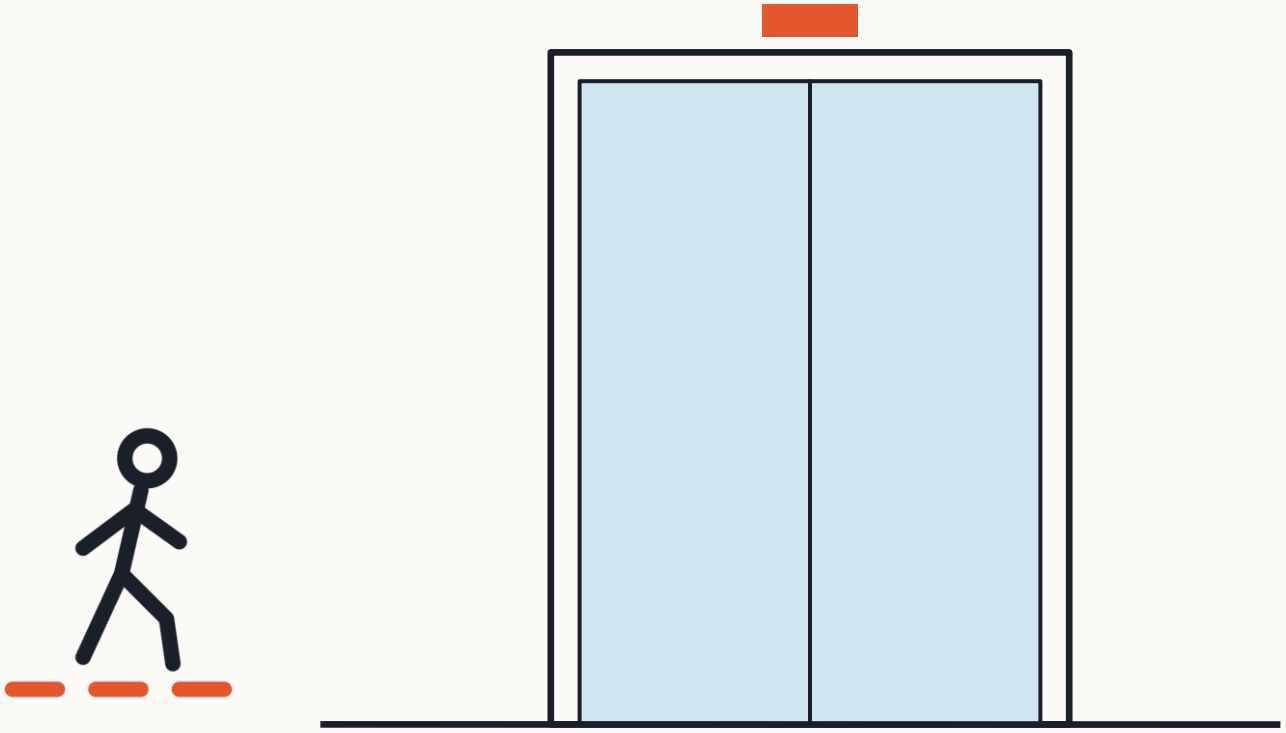
**Is the calculator intelligent?**

Yes No

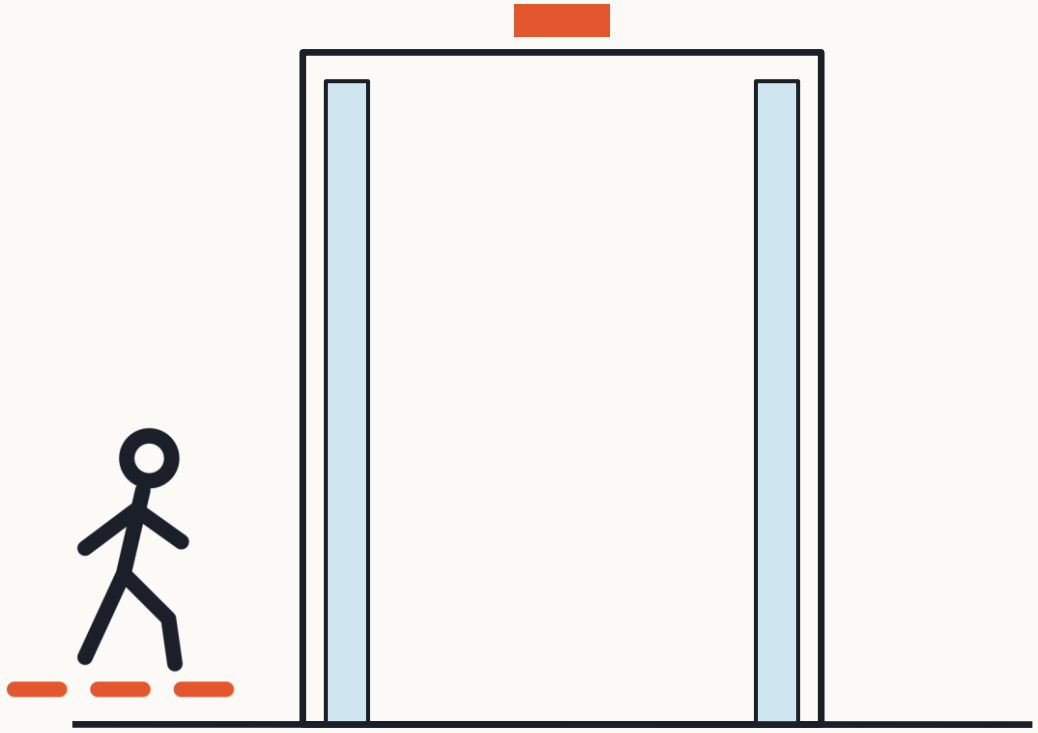
**Does speed alone mean intelligence?**

**Speed is not the same as intelligence.**

# Example 2: the automatic door



# Example 2: the automatic door



## Is that AI?

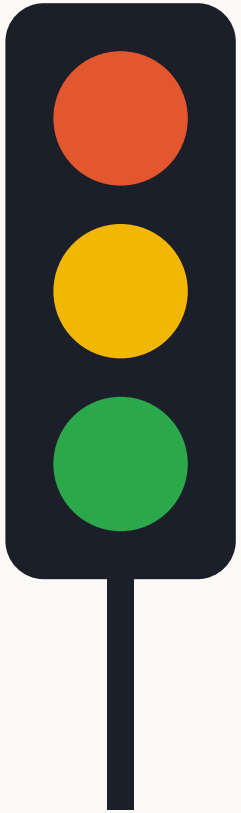
*"Yes, because it sensed me and reacted."*

*"No. It is simply following a rule:"*

```
IF something is near THEN open
```

1

# Example 3: a fixed traffic light



ONE CYCLE



green 30 s

*the same, every cycle*

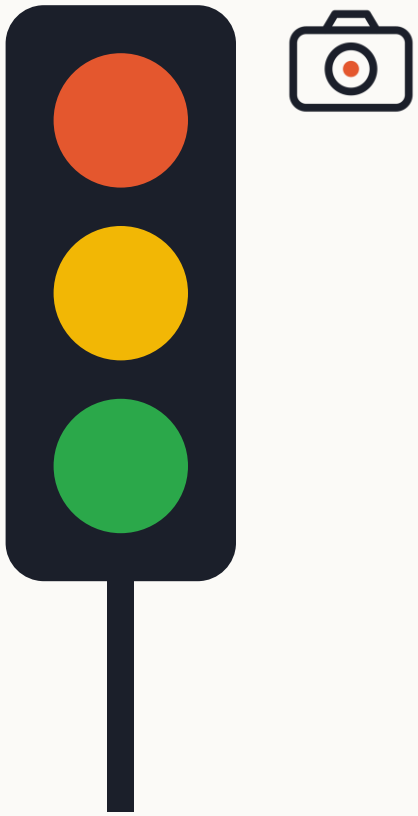


## Would you call that AI?

Information about now changes the action.

2

# Example 3: an adaptive traffic light



ONE CYCLE



green 55 s

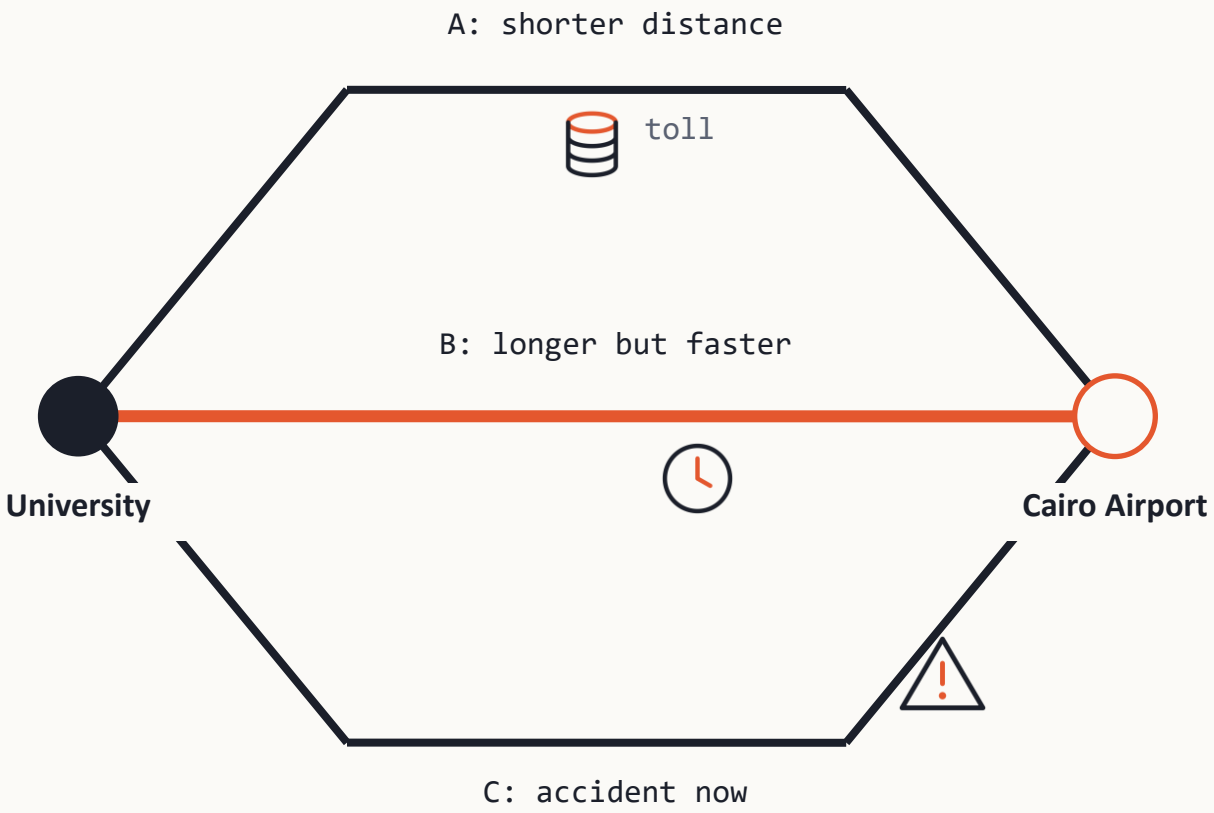
Road 1: 20 cars waiting. Road 2: almost empty.

**Would you think differently about this system?**

**What changed? The behavior.**

2

# Example 4: Google Maps




**"Take this route."**

THE SYSTEM HAS

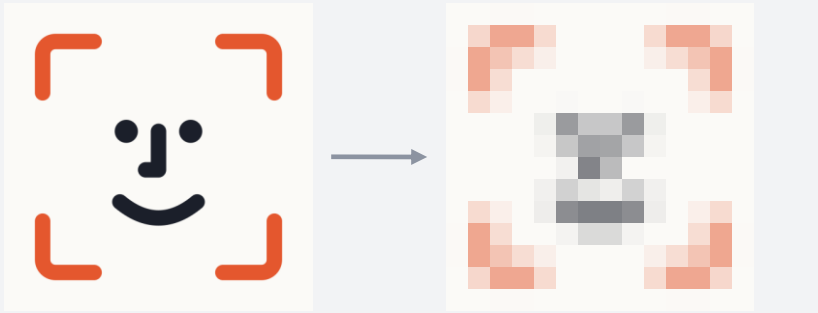
1. a current situation
2. a goal
3. several possible alternatives
4. information about the alternatives
5. a way to decide which is better

Experience can improve future choices.

3

# Face Unlock and recommendations

## RAW DATA BECOMES MEANING



pixels, numbers, colors

**Is there a face? Whose face? Does it match?**

**Unlock** | Stay locked

## PAST DATA SHAPES FUTURE CHOICES

**You**

watched
skipped
searched
completed
liked

**Your friend**

**→ estimate what you might want next**

Different history, so a different feed.

3

# So, where is the intelligence?

Is reacting enough?

Information about now changes the action.

Experience can improve future choices.

~~"Does this system use a computer?"~~

~~"Is this system fast?"~~

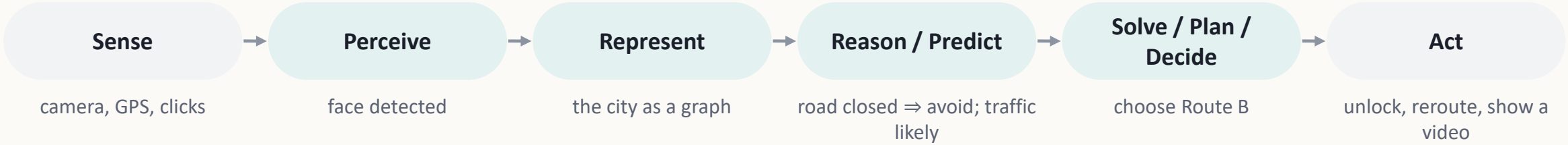
~~"Does this system automatically react?"~~

The better question is:

# Where is the intelligence?

# What makes a system intelligent?

One pipeline turns information into action.




**Learn: data or experience changes the pipeline itself.** **Adapt: behavior changes when the situation changes.**

*Language is not a separate stage: it can be the input (a question) or the output (an answer).*

**These words are related, but not the same. Five pairs show the boundaries.**

# Sensor data, perception, representation

DATA

## What did the sensor give?



Numbers: pixels.

PERCEPTION

## What did I observe?




"There is a pedestrian."

REPRESENTATION

## How do I store what I know?

```
pedestrian
  position:  12 m ahead
  moving:    left
vehicle
  speed:     40 km/h
```

Useful facts in a form the system can use.

# Representation vs reasoning

REPRESENTATION

## How do I express what I know?

```
All students registered in CS313 have a university
ID.
Mona is registered in CS313.
Road X is closed.
```

REASONING

## What can I derive from what I know?

∴ **Mona has a university ID.**

∴ **Do not select routes that use Road X.**

Neither conclusion was stored. Both were derived.

# Reasoning vs prediction

REASONING

## What follows from what I know?

All mammals are warm-blooded.
A whale is a mammal.

---

∴ A whale is warm-blooded.

PREDICTION

## What is likely to happen or be true?

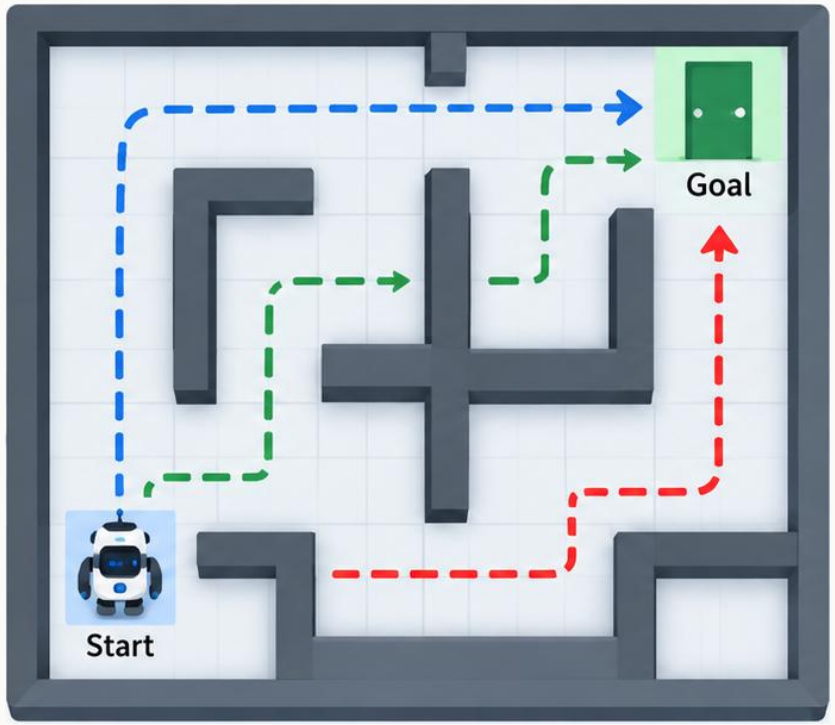


80%

chance this road becomes congested, from past patterns and current conditions

*Other predictions: is this email spam? will this customer leave? is this transaction fraud?*

# Problem solving, planning, decision making




PROBLEM SOLVING

**How can I get from where I am to where I want to be?**

Start, goal, allowed moves (no walking through walls). Find any path that works.

PLANNING

**What actions, in what order?**

Blue as a plan: up, up, up, up, then right to the goal. Swap the order and the robot hits a wall.

DECISION MAKING

**Which option should I choose now?**

Blue or red? Shortest, fewest turns, or safest? It depends on the criterion.

**A plan may contain many decisions, and every decision needs a criterion.**

# Learning vs adaptation

LEARNING

**What changes: the system itself.**

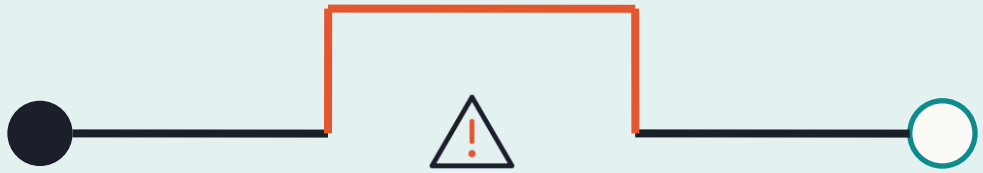
| | |
|---|---|
| You won a lottery! | spam |
| Meeting tomorrow | not spam |
| Cheap meds online | spam |

**→ from many labeled examples, the filter itself gets better**

*Seen in Part 1: recommendations*

ADAPTATION

**What changes: only the situation.**



**→ Maps reroutes. The app itself did not change.**

*Seen in Part 1: the adaptive traffic light*

THE TEST **Give it the same situation again. Same behavior → adaptation. Better behavior → learning.**

**Adaptation can come from a rule written in advance, or from what the system has learned.**

# The whole pipeline, once: the self-driving car



| | |
|---|---|
| **Sense** | camera and range data |
| **Perceive** | pedestrian, car, traffic light |
| **Represent** | pedestrian 12 m ahead, moving left |
| **Reason / Predict** | pedestrians have priority; he is likely to cross |
| **Decide** | yield instead of continuing |
| **Act** | brake gently |
| **Learn** | some of these parts were trained on data |

**There is no single "intelligence button". Several abilities work together.**

# Not every system needs every stage



**Chess engine**

does not need to recognize pedestrians



**Face recognition**

does not need to plan a route



**Spam filter**

does not need to control a robot

**What capabilities does this system need to solve its problem?**



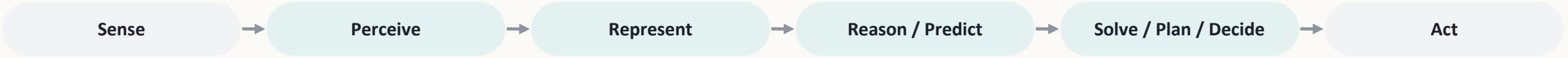
**Back to the door**

```
IF something is near THEN open
IF temperature > 30 THEN fan on
```

It senses and acts, but skips perceive, represent, reason and decide. That is automation.

**Automation ≠ AI**

# So: where is the intelligence?




## AI is not one algorithm.

## It is not one model.

## It is not Machine Learning alone.

## And it is definitely not only ChatGPT.

**AI is a broad field: building systems that perform tasks requiring different forms of intelligent behavior.**

# Do not think of AI as one technology

ASK

## What technique does Computer Science use?

- algorithms
- databases
- networks
- operating systems
- security
- software engineering

There is no single answer. AI is the same: a large field with different problems and approaches.

A better question:

## What kind of problem are we trying to solve?



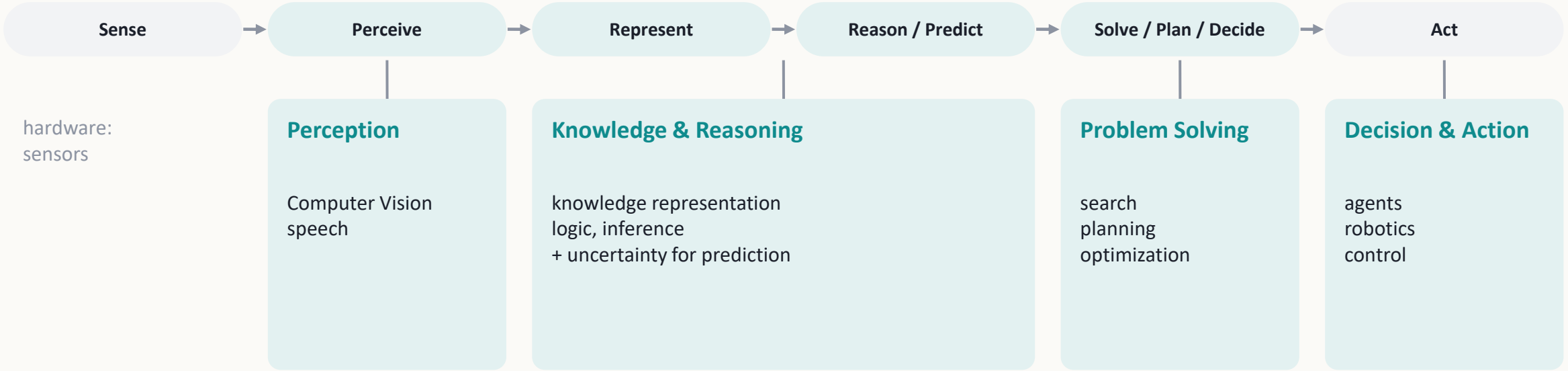
route through a maze vs  recognize a face

**Same technique?**



**Different AI problems require different approaches.**

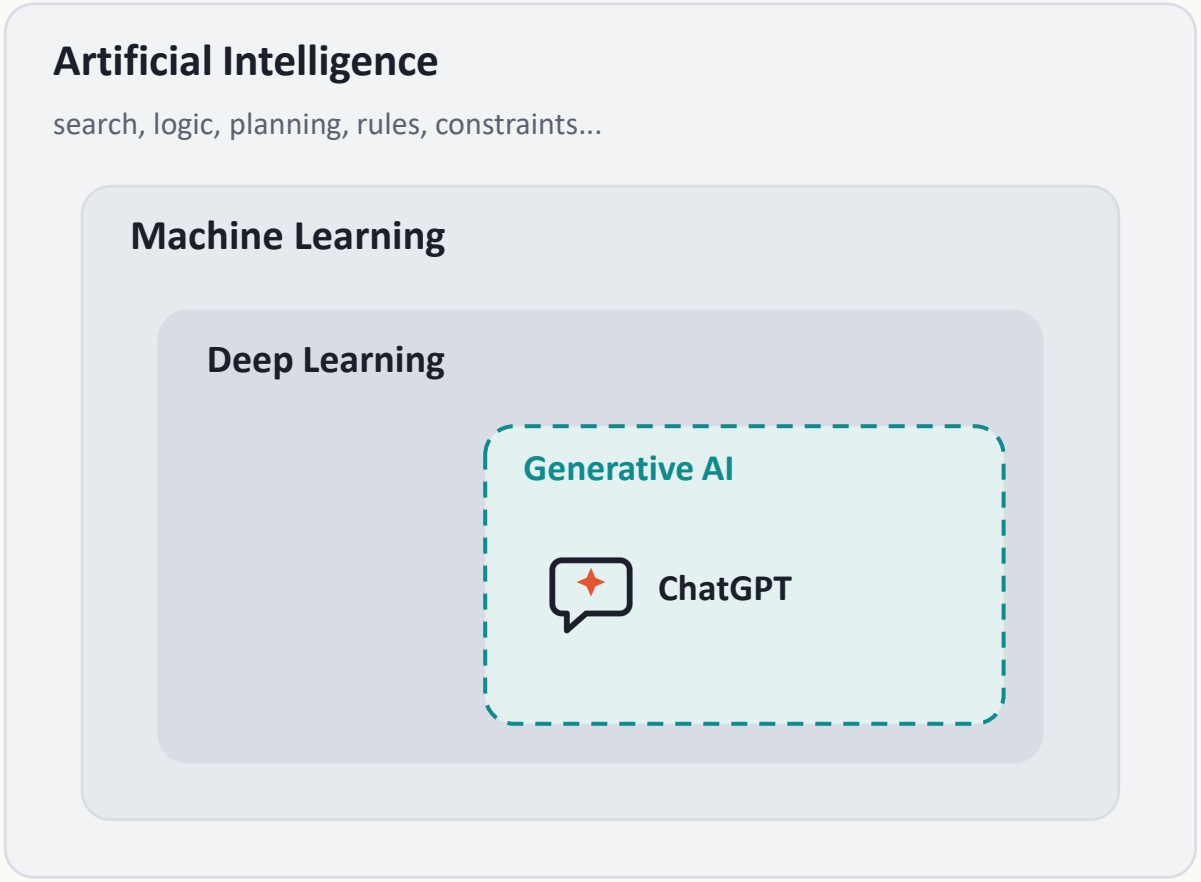
# The map of AI is the pipeline, relabelled




**Learning: Machine Learning, and inside it Deep Learning. It can improve any stage.**

**Language: Natural Language Processing and language models, at the input and at the output.**

# AI, ML, DL, and where ChatGPT sits




| | |
|---|---|
| **AI** | The broad field. |
| **ML** | Learns patterns from data instead of only hand-written rules. |
| **DL** | ML with multilayer neural networks. |
| **Generative AI** | Systems that produce new content: text, images, audio, code. |

**ChatGPT: an LLM-based system in NLP and Generative AI. One system inside the field, not the field.**

# Which parts of AI does this system use?



## Face Unlock

Perception (Computer Vision), representation, a learned model, decision.



## Chess engine

Problem solving, search, decision. Can be AI with no Machine Learning at all.

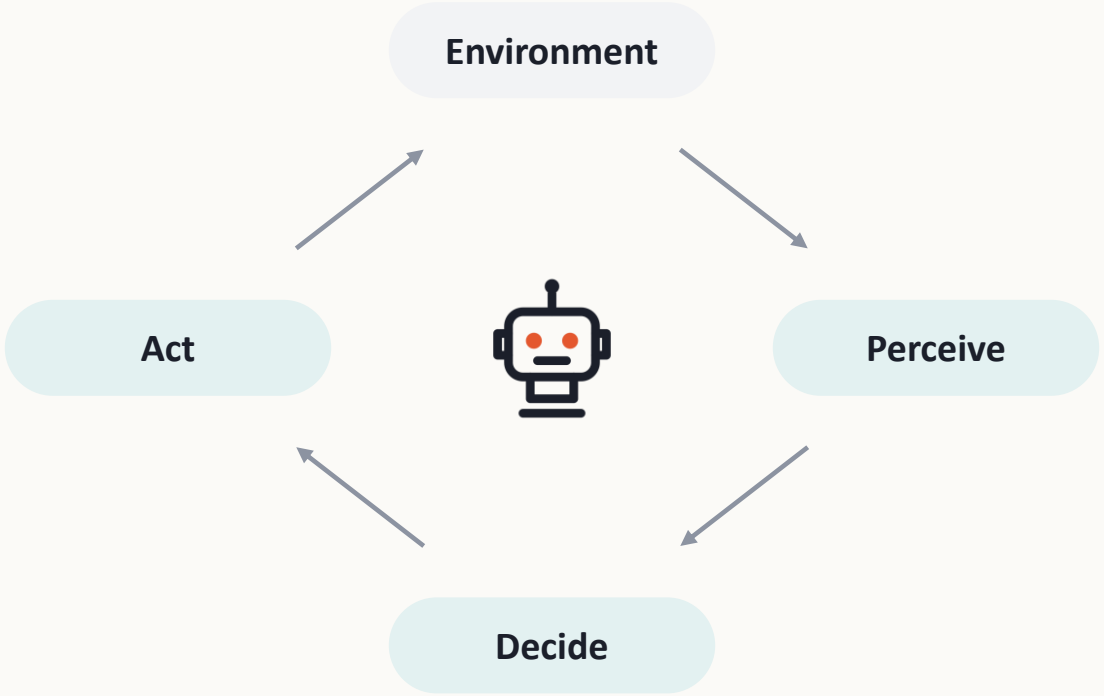


## Google Translate

Language (NLP), Machine Learning, Deep Learning; speech if the input is spoken.

*Do not ask "which box is it?" Ask "which areas does it use?"*

# Who runs the pipeline?




Close the pipeline into a loop with the world.

AGENT

**An entity that perceives its environment and takes actions within that environment.**

Observe → Decide → Act, again and again. The action changes the environment, and the agent perceives again.

# Agents are not only robots

PHYSICAL

## Robot vacuum

in: senses walls, obstacles, dirt
out: moves, turns, cleans

SOFTWARE

## Email agent

in: receives messages
out: moves to spam, allows, flags

GAME

## Chess program

in: board, opponent moves
out: selects a move

SENSORS

**information comes in**

camera, microphone, GPS; for software: messages, files, API data

ACTUATORS

**action goes out**

wheels, motors, brakes; or digital: send a message, write a file

GOAL

**which actions are useful**

reach the exit, win the game, keep the floor clean

Rationality, agent types and PEAS: Section 2.

PART 5

# What does the rest of CS313 look like?

Imagine I say: "Okay. Build an agent." Every question that appears is one stop in this course.

PHASE 1

# Foundations

**What makes a system intelligent?**

**01** **AI Around Us**

Example: Today: from calculators to self-driving cars.

**Who perceives and acts, and how should it act?**

**02** **Intelligent Agents**

Example: Robots, cars and software agents.

**What exactly is the problem: states, actions, goal?**

**03** **Problem Formulation**

Example: A robot in Room A must reach Room D; each room is a state.

→ *Once we know who acts and what the problem is, we need a way to find solutions.*

PHASE 2

# Solving by Search

**Given a start, actions and a goal, how do we find a solution?**

## 04 Search

Example: Mazes, route finding, puzzles.

**If the space is huge, can extra information guide the search?**

## 05 Searching Smarter

Example: Distance-to-goal estimates; later Greedy and A*.

**What if the answer is an assignment that must satisfy many rules?**

## 06 Constraints

Example: Timetables, Sudoku, N-Queens.

→ *Some problems are not paths but assignments, and the system also needs knowledge about the world.*

PHASE 3

# Knowledge & Reasoning

**How can the system store facts, relations and rules?**

## 07 Knowledge

Example: A family tree: who is whose parent.

**Once knowledge is represented, what can the system conclude?**

## 08 Reasoning

Example: A parent of a parent is a grandparent.

**What if information is incomplete, noisy or uncertain?**

## 09 Uncertainty

Example: Diagnosis, traffic, pedestrian behavior.

→ *If writing all the knowledge and rules by hand is impractical, can the system learn them?*

PHASE 4

# Learning

**What if writing all the rules by hand becomes impractical?**

**10** **Learning from Data**

Example: Spam filters, vision, recommendations.

**How do ML, DL, NLP, Transformers, LLMs and GenAI connect?**

**11** **Modern AI**

Example: A map for orientation, not full depth.

**How are real AI systems built from several ideas?**

**12** **Bringing It Together**

Example: An integrated case study and project presentations.

→ *Real systems combine every idea from the course.*

# Three levels of depth

CORE

**You need to understand it deeply.**

Agent foundations, problem formulation, state space, search, heuristics, CSP, knowledge representation, reasoning, uncertainty.

MAP

**Know what it is, why it exists, where it belongs.**

Deep Learning, Computer Vision, NLP, LLMs, Generative AI.

EXTEND

**Know it exists and where to explore further.**

Genetic Algorithms, metaheuristics, Reinforcement Learning, advanced planning, multi-agent systems, advanced Bayesian models.



# How we will study this course

1

## Problem before terminology

Before an algorithm: what problem created the need for it?

2

## Tree before branch

Before a technique: where does it belong in AI?

3

## Reason before recipe

Before implementation: why does it work?

# Where is the intelligence? Trace one system



## Google Maps

| | |
|---|---|
| **Sense** | GPS location, live traffic data |
| **Perceive** | congestion on a road |
| **Represent** | roads as a graph, travel times as costs |
| **Reason / Predict** | a closed road is avoided; traffic is likely to grow |
| **Solve / Plan / Decide** | find routes, order the segments, choose the fastest |
| **Act** | show the route |
| **Learn / Adapt** | learn traffic patterns; reroute after a closure |



**The intelligence is not the map. It is what the system does with information.**

# Exit questions

| | | |
|---|---|---|
| 1 | A camera captures an image. Is that perception? | **Not by itself. Interpreting it is.** |
| 2 | Road A is closed, so do not use it. Reasoning or prediction? | **Reasoning** |
| 3 | Road A may be congested in ten minutes. Reasoning or prediction? | **Prediction** |
| 4 | Choose Route A or B by the fastest time. | **Decision making** |
| 5 | Maps reroutes right after a road closes. | **Adaptation, not learning** |

# Your task: Spot the Intelligence

Choose one real system you believe uses AI (not Google Maps) and analyze it with today's framework.

1. What problem does the system solve?
2. What information does it receive?
3. What does it need to represent internally?
4. Which intelligent capabilities are involved? (only the ones that genuinely apply)
5. What output or action does it produce?
6. Does it appear to use Machine Learning? If yes, what is learned? If no, what other AI method?
7. **Where exactly is the intelligence?**

**LENGTH**

About one page, or one structured diagram with a short explanation.

**SUBMISSION**

```
Due: Thursday 1/10
Format: on the Telegram group
```

*Analysis, not a long report.*

# Weak vs strong answers

WEAK

*"Google Maps is AI because it uses Machine Learning and gives smart routes."* *It explains almost nothing.*

STRONGER

"Google Maps solves a navigation problem. It receives location, destination, road-network information and traffic. It represents possible routes and their estimated costs. Search or route-planning methods find paths; prediction estimates travel time. It compares alternatives, recommends a route, and adapts when traffic changes. The intelligence is not displaying a map; it is representing the problem, evaluating alternatives, predicting conditions and selecting a route."

**Not more words. Better analysis.**

WHERE WE STAND

An agent perceives an environment, chooses actions, and tries to achieve some objective. But we left many questions unanswered.

**What makes an agent rational?**

**How do we formally describe what an agent perceives?**

**How do we describe its task?**

**What kind of environment is it operating in?**

**Should every agent behave in the same way?**

NEXT 02 Intelligent Agents →




Questions? Ask now, or on the Telegram group.

**Dr. Ahmed Sleem**

Eng. Omar Khaled · Eng. Nagwa Mohamed

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